OPEN SUBSTACKS CORRESPOND BIJECTIVELY TO OPEN SUBSETS OF THE POINT SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2065-2071 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 2073-2082, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 37 of queue_stacks-properties.json (pb=420, stmt 2065-2071, proof 2073-2082).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04T5: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2082–2082. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-points-presentation: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1102–1108. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-substacks-presentation: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2026–2039. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-topology-points: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1155–1164. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. The rule $\mathcal{U} \mapsto |\mathcal{U}|$ defines an inclusion preserving bijection between open substacks of $\mathcal{X}$ and open subsets of $|\mathcal{X}|$.*

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose a presentation $[U/R] \to \mathcal{X}$, see Algebraic Stacks, Lemma 04T5. By Lemma lemma-substacks-presentation we see that open substacks correspond to R-invariant open subschemes of U. On the other hand Lemmas lemma-points-presentation and lemma-topology-points guarantee these correspond bijectively to open subsets of $|\mathcal{X}|$. $\square$